I. VEHICLE RENTAL SYSTEM SPECIFICATION

The Vehicle Rental store needs to build an online vehicle rental system with the following core functionalities:

- ***Browse and Search Vehicles:***
  - + Customers can view vehicles by **vehicle type** (motorbikes, 4-seater cars, 7-seater cars, electric vehicles, etc.). The vehicle list can be sorted (ascending or descending) by criteria such as rental price, discount, and rental count.
  - + Customers can search for vehicles by entering keywords. The system will return a list of vehicles whose name or model contains the keyword. The vehicle list can be sorted (ascending or descending) by rental price, discount, and rental count.
- ***Select and Book Vehicles:***
  - + Each **vehicle** has a vehicle name, unique vehicle code, license plate, illustrative images, summary and detailed descriptions (make, seat capacity, fuel type, transmission), daily rental price, availability status (available or rented), and view count. Each vehicle belongs to a specific **vehicle category**.
  - + When browsing or searching for vehicles, for each item in the list, customers can *add the selected vehicle to the current rental cart*, or *view detailed information about the vehicle*. While viewing details, customers can also *add the vehicle to the rental cart.*
  - + During the selection process, customers can *view the list ofselected vehicles in the cart, update the rental cart* (remove vehicles, update rental duration and vehicle quantities).
  - + Customers select the **booking function** after finishing their vehicle selections to officially place a rental reservation.
  - + **Vehicle Booking Process:**
    - The system will prompt the customer to **log in** if not logged in, or **register a new account** if they do not have an account yet.
    - Customers enter the **recipient/driver information** (full name, address, phone number, driver's license number). Note that the driver/recipient can be different from the buyer (e.g., booking for a colleague or relative).
    - If the rental order value is **2,000,000 VND** or more, delivery is free. Otherwise, the system calculates a **delivery fee** depending on the **delivery area** (based on recipient address down to ward level) and selected delivery method. The delivery fee (ifany) is added to the invoice total.
    - Customers can choose cash payment upon vehicle handover or online payment. *Ifonline payment is chosen, the system automatically connects to the Online Payment Service Gateway*.

SPECIFICATION (DRAFT)

- Upon successful booking, the system will **record the rental order**, including vehicle items, *quantity* and corresponding *unit price*, renter and recipient info, delivery method, total order value, timestamp, and payment method.
- Ifan email is provided, the system will **send a booking confirmation email** with full order details.

- ***Customer Registration & Profile Updates:***
  - + Customers can register an account in the system.
  - + Customer information includes: full name, date of birth, gender, address, phone number, password, email address (used as login username, order notifications, and password recovery).
  - + Customers can update all their profile information except their username (email).
  - + Forgotten Password case: Customers use the password recovery function; the system generates a new password and sends it to their email.
- ***View Placed Rental Orders:***
  - + Customers can review placed rental orders and their status (New, Confirmed, Delivering, In-Rental, Completed, Cancelled).
  - + Customers can cancel an order if its status is "New".
- ***StaffOperations:***
  - + View rental orders and update processing status.
  - + Perform vehicle handover and vehicle return inspection procedures.
- ***System Administration:*** Administrators have the following functions:
  - + Manage vehicles and vehicle categories
  - + Manage users (Customers & Staff)
  - + Manage rental orders

II. GLOSSARY

1. **Discount:** Calculated as a percentage (%) ofthe current rental price.
2. **Rental Cart:** List ofvehicles selected by the customer prior to order confirmation/deposit.
3. **Vehicle Category:** Classification ofvehicles (Motorbikes, 4-seater cars, 7-seater cars, Electric vehicles, etc.).

SPECIFICATION (DRAFT)

III. Supplementary Specification

Usability

Responsive interface that adapts to screen size.

Reliability

The system must operate 24/7 with a reliability of > 99.5%.

Performance

The system must handle 2,000 concurrent users with zero errors.

The system must scale to handle over 100,000 customers, 10,000 vehicles, and 1,000,000 bookings without degrading response times

Supportability

None

Security

Customers can only view their own information and orders. Data must be encrypted at rest and viewable only by authorized staff. Use parameterized queries (SQL injection prevention), output encoding (XSS prevention), and CSRF tokens on all state-changing requests.

Design Constraints

HTML5, CSS3, JavaScript front end; server-side language of the team's choice and MySQL database management system

IV. Data Flow Diagram

1) DFD Level 0 (context diagram)

The whole system is one process. It exchanges data with three external entities: the Customer, the Admin/staff and the Payment gateway.

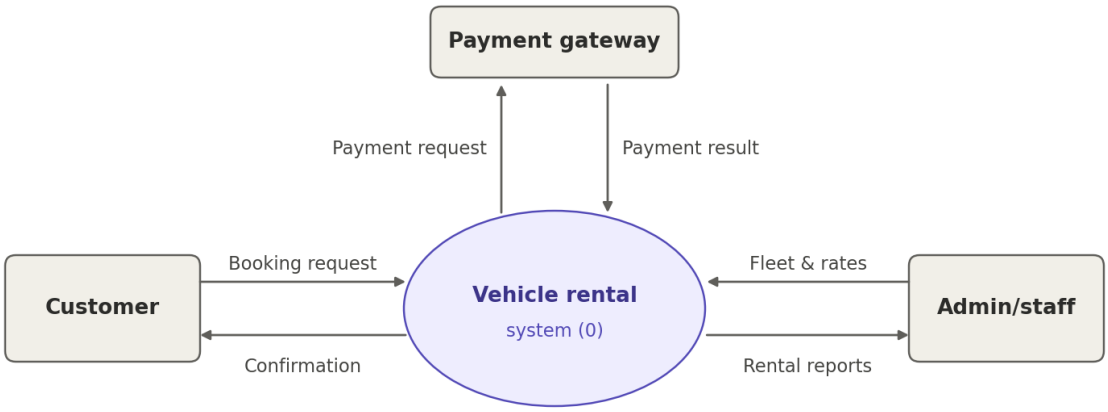


Figure 1. DFD Level 0 – Vehicle Rental System

Data flow	From	To	Data carried
Booking request	Customer	System (0)	Registration and login details, search criteria, pickup and return dates, vehicle choice, payment choice, return of the vehicle
Confirmation	System (0)	Customer	Search results, booking confirmation, invoice, rental status, email notice
Fleet & rates	Admin/staff	System (0)	Vehicle details, daily rates, fees, rental handover and return inspection, account administration
Rental reports	System (0)	Admin/staff	Booking lists, utilisation and revenue reports, vehicle status
Payment request	System (0)	Payment gateway	Amount, booking reference, customer payment redirect
Payment result	Payment gateway	System (0)	Success or failure, transaction reference

2) DFD Level 1

Process 0 is split into five sub-processes and three data stores. Arrows are double-headed because each link carries a request in one direction and a response in the other.

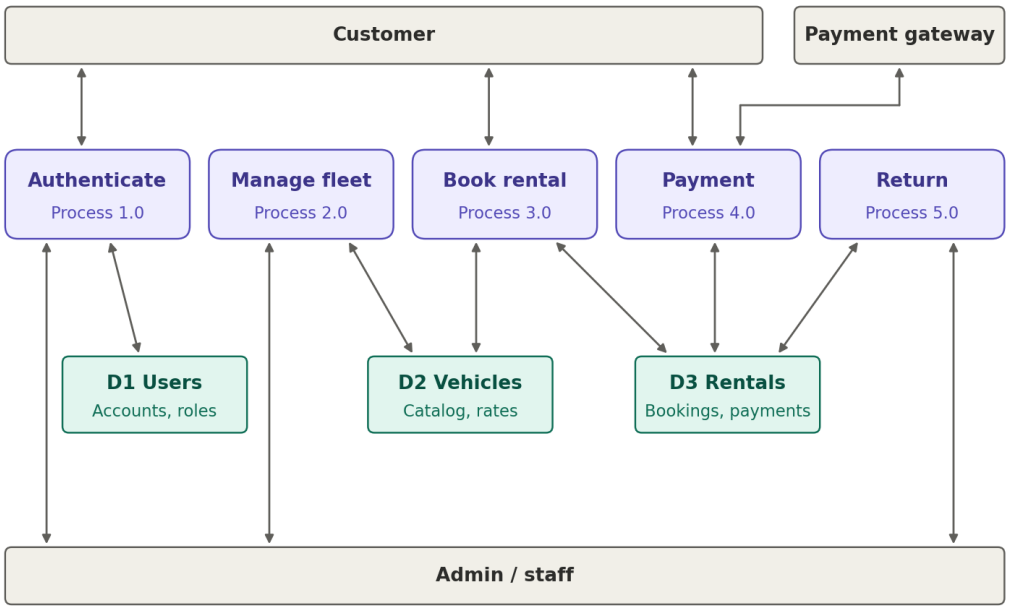


Figure 2. DFD Level 1 – processes, data stores and external entities

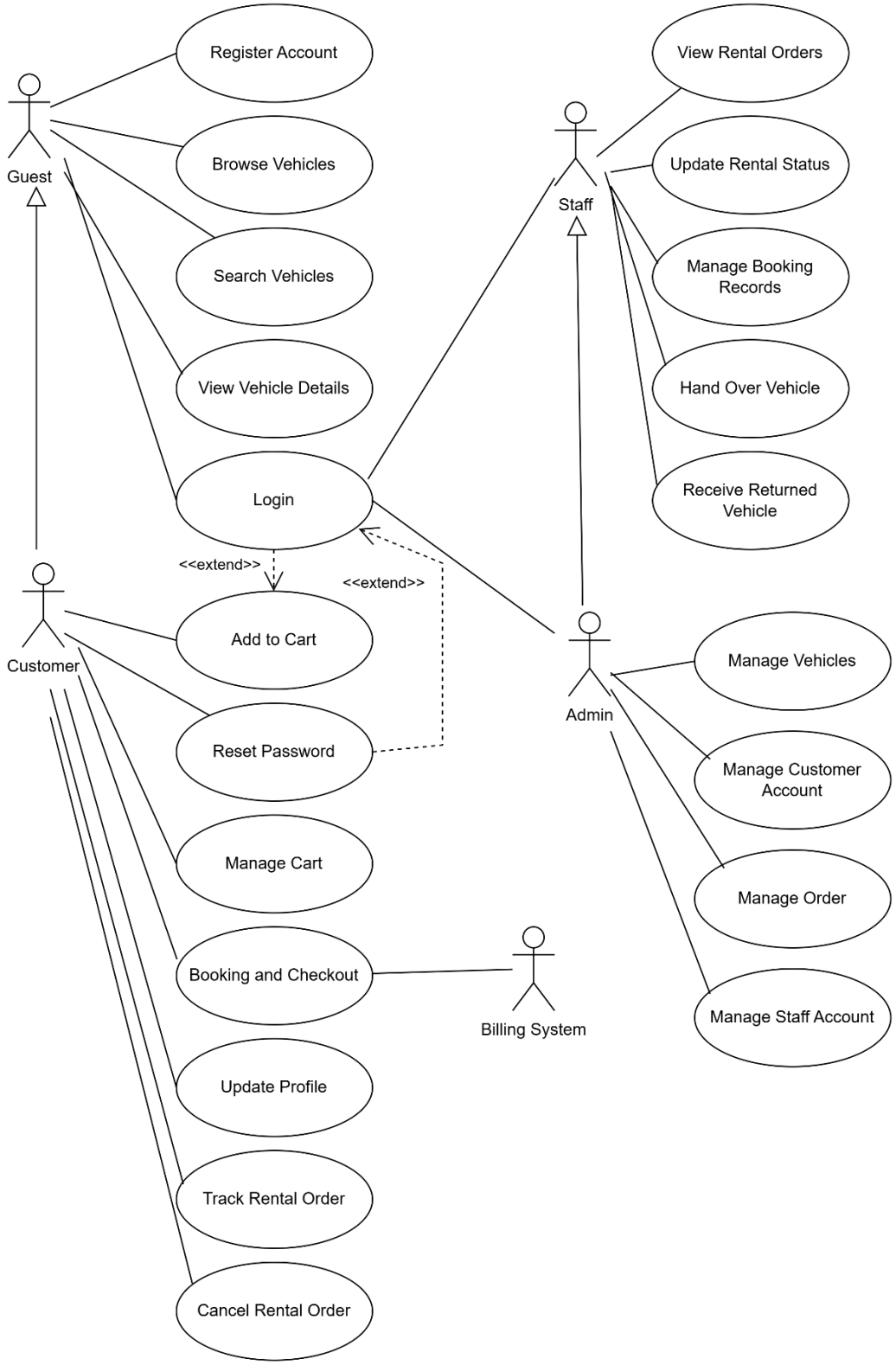
Process	Name	Inputs	Outputs	Data stores
1.0	Authenticate	Registration details and login credentials from Customer and Admin/staff	Login result, session, role	D1 Users (read/write)
2.0	Manage fleet	Vehicle details, rates and status changes from Admin/staff	Updated vehicle list and confirmation	D2 Vehicles (read/write)
3.0	Book rental	Search criteria, dates and vehicle choice from Customer; vehicle catalog from D2	Search results, booking record, confirmation; availability checked against dates already booked in D3	D2 (read), D3 (read/write)
4.0	Payment	Payment choice from Customer; booking total from D3; result from Payment gateway	Payment request to gateway, invoice and receipt to Customer, payment status	D3 (read/write)
5.0	Return	Return and inspection details from Admin/staff	Final charges, rental status 'closed', return report	D3 (read/write)

3) Consistency and assumptions

- Customer, Admin/staff and Payment gateway appear in both Level 0 and Level 1.
- The Customer flows of Level 0 are covered by processes 1.0, 3.0 and 4.0; the Admin/staff flows by 1.0, 2.0 and 5.0; the Payment gateway flows by 4.0 only.
- Confirmation emails are sent inside processes 3.0 and 4.0, so no separate email entity is drawn.
- D3 Rentals holds both bookings and payments to keep the diagram simple. In the database design they can be split into two tables.
- Vehicle availability is derived from the dates already booked in D3, so process 5.0 does not need to write to D2.

V . Use Case Diagram

1) Overview Diagram



1.1) List of Actors

No	Use-case	Description
1	Guest	Unauthenticated users can only browse public pages
2	Customer	Authenticated users with a registered account
3	Admin	A user responsible for managing the system,configuring settings, and monitoring logs
4	Billing System	Payment Gateway

1.2) List of Use-case

No	Use-case	Description	Group Notes
1	Register Account	Register a new account	N/A
2	Browse Vehicles	Browse available vehicles	N/A
3	Search Vehicles	Search for vehicles	N/A
4	View Vehicle Details	View detailed vehicle information	N/A
5	Login	Log in to the system	N/A
6	Add to Cart	Add vehicles to the rental cart	N/A
7	Reset Password	Reset a forgotten password	N/A
8	Manage Cart	Manage the rental cart	N/A
9	Booking and Checkout	Book vehicles and make payment	N/A
10	Update Profile	Update personal information	N/A
11	Track Rental Order	Track rental order status	N/A
12	Cancel Rental Order	Cancel a rental order	N/A
13	View Rental Orders	View the list of rental orders	N/A

14	Update Rental Status	Update rental order status	N/A
15	Manage Booking Records	Manage booking records	N/A
16	Hand Over Vehicle	Hand over the vehicle to the customer	N/A
17	Receive Returned Vehicle	Receive the returned vehicle	N/A
18	Manage Vehicles	Manage vehicles	N/A
19	Manage Customer Account	Manage customer accounts	N/A
20	Manage Order	Manage rental orders	N/A
21	Manage Staff Account	Manage staff accounts	N/A

USE CASE SPECIFICATION – VEHICLE RENTAL SYSTEM

1) Use Case: Register Account

1.1) Brief Description

This use case allows visitors to create a new account to access the vehicle rental system.

1.2) Flow of Events

1.2.1) Basic Flow

This use case begins when a visitor selects the Register Account function.

1. The system displays the registration form.
2. The visitor enters their full name, date of birth, gender, address, phone number, password, and email.
3. The visitor submits the registration request.
4. The system validates the entered information.
5. If the information is valid, the system saves the new account.
6. The system displays a registration confirmation message.

1.2.2) Alternative Flows

- If the email already exists, the system displays an error message and asks the visitor to enter another email.
- If required information is missing or invalid, the system displays an error message and asks the visitor to correct it.

1.3) Special Requirements

No special requirements are specified.

1.4) Pre-conditions

The visitor does not have an account or has not logged in to the system.

1.5) Post-conditions

If successful, a new account is created in the system. Otherwise, the account data remains unchanged.

1.6) Extension Points

None.

2) Use Case: Browse Vehicles

2.1) Brief Description

This use case allows users to browse available vehicles by category, such as motorbikes, 4-seater cars, 7-seater cars, and electric vehicles.

2.2) Flow of Events

2.2.1) Basic Flow

This use case begins when a user accesses the Browse Vehicles function.

1. The system displays the available vehicle categories.
2. The user selects a vehicle category.
3. The system displays the vehicles belonging to the selected category.
4. The user can sort the vehicles by rental price, discount, or rental count.
5. The system displays the vehicle list according to the selected sorting criteria.

2.2.2) Alternative Flows

- If no vehicles are available in the selected category, the system displays a notification indicating that no vehicles were found.

2.3) Special Requirements

No special requirements are specified.

2.4) Pre-conditions

The system is operational and contains vehicle information.

2.5) Post-conditions

The vehicle list is displayed according to the selected category and sorting criteria.

2.6) Extension Points

- Add to Cart: The user selects a vehicle to add to the rental cart.
- View Vehicle Details: The user selects a vehicle to view its detailed information.

3) Use Case: Search Vehicles

3.1) Brief Description

This use case allows users to search for vehicles by entering keywords related to the vehicle name or model.

3.2) Flow of Events

3.2.1) Basic Flow

This use case begins when a user selects the Search Vehicles function.

1. The system displays the search input field.
2. The user enters the name or model of the vehicle.
3. The user submits the search request.
4. The system searches for vehicles matching the entered keywords.
5. The system displays the search results.
6. The user can sort the results by rental price, discount, or rental count.
7. The system displays the results according to the selected sorting criteria.

3.2.2) Alternative Flows

- If no matching vehicles are found, the system displays a notification indicating that there are no search results.
- If the search keyword is invalid, the system asks the user to enter it again.

3.3) Special Requirements

No special requirements are specified.

3.4) Pre-conditions

The system is operational and contains vehicle data for searching.

3.5) Post-conditions

The search results are displayed according to the entered keywords and selected sorting criteria.

3.6) Extension Points

- Add to Cart: The user selects a vehicle to add to the rental cart.
- View Vehicle Details: The user selects a vehicle to view its detailed information.

4) Use Case: View Vehicle Details

4.1) Brief Description

This use case allows users to view detailed information about a selected vehicle.

4.2) Flow of Events

4.2.1) Basic Flow

This use case begins when a user selects a vehicle from the list.

1. The system receives the request to view vehicle details.
2. The system retrieves the information of the selected vehicle.
3. The system displays detailed information, including vehicle name, unique code, license plate, images, short description, detailed description, make, seat capacity, fuel type, transmission, daily rental price, availability, and category.
4. The user views the vehicle information.

4.2.2) Alternative Flows

- If the vehicle no longer exists or cannot be found, the system displays a notification indicating that the vehicle information cannot be displayed.

4.3) Special Requirements

No special requirements are specified.

4.4) Pre-conditions

The user is accessing the system, and the selected vehicle exists in the system.

4.5) Post-conditions

The detailed information of the selected vehicle is displayed to the user.

4.6) Extension Points

- Add to Cart: The user selects the option to add the vehicle to the rental cart.

5) Use Case: Login

5.1) Brief Description

This use case describes how users log in to the system using their registered accounts.

5.2) Flow of Events

5.2.1) Basic Flow

This use case begins when a user selects the Login function.

1. The system asks the user to enter their email and password.
2. The user enters their login information and submits the request.
3. The system verifies the email and password.
4. If the information is correct, the system allows the user to log in.
5. The system redirects the user to the interface corresponding to their account permissions.

5.2.2) Alternative Flows

- If the email or password is incorrect, the system displays an error message and asks the user to try again.
- If the user cancels the login process, the use case ends.

5.3) Special Requirements

No special requirements are specified.

5.4) Pre-conditions

The user has a registered account and is not currently logged in.

5.5) Post-conditions

If successful, the user is authenticated and can access functions according to their permissions. Otherwise, the login status remains unchanged.

5.6) Extension Points

- Reset Password: The user accesses the password recovery function when they forget their password.

6) Use Case: Add to Cart

6.1) Brief Description

This use case allows users to add vehicles to the rental cart before proceeding with the booking process.

6.2) Flow of Events

6.2.1) Basic Flow

This use case begins when a user selects the Add to Cart function.

1. The user selects a vehicle from the vehicle list or detail page.
2. The system receives the request to add the vehicle.
3. The system checks the vehicle information and availability.
4. If the vehicle can be added, the system updates the rental cart.
5. The system displays a notification confirming the result.

6.2.2) Alternative Flows

- If the vehicle is no longer available, the system displays a notification and does not add the vehicle to the rental cart.
- If the vehicle information is invalid, the system displays an error message.

6.3) Special Requirements

No special requirements are specified.

6.4) Pre-conditions

The user is accessing the system and has selected a vehicle.

6.5) Post-conditions

If successful, the selected vehicle is added to the rental cart. Otherwise, the cart remains unchanged.

6.6) Extension Points

- Manage Cart: The user accesses the rental cart to view and manage its contents.

7) Use Case: Reset Password

7.1) Brief Description

This use case allows users to recover their passwords when they forget their login credentials.

7.2) Flow of Events

7.2.1) Basic Flow

This use case begins when a user selects the Reset Password function.

1. The system asks the user to enter their registered email address.
2. The user enters the email address and submits the recovery request.
3. The system checks whether the email exists in the system.
4. If the email is valid, the system generates a new password.
5. The system sends the new password to the user's registered email address.

6. The system displays a notification about the password recovery result.

7.2.2) Alternative Flows

- If the email does not exist, the system displays an error message and asks the user to check the email address.

7.3) Special Requirements

The new password must be sent to the user's registered email address.

7.4) Pre-conditions

The user is not logged in and has an account registered in the system.

7.5) Post-conditions

If successful, the account password is changed. Otherwise, the account information remains unchanged.

7.6) Extension Points

None.

8) Use Case: Manage Cart

8.1) Brief Description

This use case allows users to view and manage the vehicles in their rental cart, including updating rental duration or quantity and removing vehicles.

8.2) Flow of Events

8.2.1) Basic Flow

This use case begins when a user selects the Manage Cart function.

1. The system displays the list of vehicles in the rental cart.
2. The system displays the vehicle information and selected rental duration.
3. The user can update the rental duration or quantity.
4. The user submits the cart update request.
5. The system validates the information and updates the rental cart.
6. If the user chooses to remove a vehicle, the system asks for confirmation.
7. The user confirms the removal.
8. The system removes the vehicle and updates the cart information.

8.2.2) Alternative Flows

- If the rental duration or quantity is invalid, the system displays an error message and asks the user to make corrections.
- If the user does not confirm the removal, the system keeps the vehicle in the cart.
- If the cart is empty, the system displays an empty cart notification.

8.3) Special Requirements

No special requirements are specified.

8.4) Pre-conditions

The user has added at least one vehicle to the rental cart.

8.5) Post-conditions

If successful, the rental cart is updated according to the user's selections.

8.6) Extension Points

- Booking and Checkout: The user proceeds to the vehicle booking and payment process.

9) Use Case: Booking and Checkout

9.1) Brief Description

This use case allows customers to book vehicles, provide recipient and driver information, calculate delivery fees, and complete payment.

9.2) Flow of Events

9.2.1) Basic Flow

This use case begins when a customer selects the Booking and Checkout function.

1. The system checks the customer's login status.
2. If the customer is not logged in, the system asks them to log in or register an account.
3. The system displays the selected vehicles and asks the customer to provide recipient and driver information, including full name, address, phone number, and driving license number.
4. The customer enters the required information. The recipient may be different from the person placing the order.
5. The system validates the information and calculates the delivery fee.
6. If the total order value is at least 2,000,000 VND, delivery is free. Otherwise, the delivery fee is calculated based on the ward and delivery method.
7. The system displays the total booking cost.
8. The customer confirms the booking information.
9. The system asks the customer to select a payment method: COD or Online Payment.
10. If the customer selects Online Payment, the system connects to the Online Payment Service Gateway.
11. If the payment is successful or the customer selects COD, the system records the rental order.
12. The system sends a booking confirmation email if the customer provides an email address.

9.2.2) Alternative Flows

- If the customer is not logged in, the system redirects them to Login or Register Account.
- If the recipient or driver information is invalid, the system displays an error message and asks the customer to enter the information again.
- If online payment fails, the system displays an error message and allows the customer to retry the payment.
- If the customer cancels the process, the system does not record the rental order.

9.3) Special Requirements

- The system must support two payment methods: COD and Online Payment.
- The system must connect to the Online Payment Service Gateway when the customer selects Online Payment.
- The system must apply the free delivery policy to orders valued at 2,000,000 VND or more.

9.4) Pre-conditions

The customer has selected vehicles in the rental cart and must log in before completing the booking.

9.5) Post-conditions

If successful, the rental order is saved in the system with the New status. A confirmation email is sent if the customer provides an email address. Otherwise, the rental order is not recorded.

9.6) Extension Points

- Login: The customer logs in if they do not have an active session.
- Register Account: The customer creates an account if they do not already have one.
- Online Payment Service Gateway: The system connects to the payment gateway when the customer selects Online Payment.

10) Use Case: Update Profile

10.1) Brief Description

This use case allows logged-in users to update their personal account information.

10.2) Flow of Events

10.2.1) Basic Flow

This use case begins when a user selects the Update Profile function.

1. The system displays the user's current personal information.
2. The user edits the information that can be updated.
3. The user submits the update request.
4. The system validates the information.
5. If the information is valid, the system saves the changes.
6. The system displays a notification about the update result.

10.2.2) Alternative Flows

- If the entered information is invalid, the system displays an error message and asks the user to correct it.
- If the user cancels the operation, the system does not save the changes.

10.3) Special Requirements

Users are not allowed to change their registered username and email address.

10.4) Pre-conditions

The user is logged in to the system.

10.5) Post-conditions

If successful, the user's personal information is updated. Otherwise, the information remains unchanged.

10.6) Extension Points

None.

11) Use Case: Track Rental Order

11.1) Brief Description

This use case allows customers to track the status and progress of their rental orders.

11.2) Flow of Events

11.2.1) Basic Flow

This use case begins when a customer selects the Track Rental Order function.

1. The system retrieves the customer's rental orders.
2. The system displays the list of rental orders and their corresponding statuses.
3. The customer selects a rental order to view.
4. The system displays the detailed information and processing progress of the selected order.

11.2.2) Alternative Flows

- If the customer has no rental orders, the system displays a notification indicating that no orders are available.
- If the selected rental order cannot be found, the system displays an error message.

11.3) Special Requirements

The system supports the following rental order statuses: New, Confirmed, Delivering, In-Rental, Completed, and Cancelled.

11.4) Pre-conditions

The customer is logged in and has at least one rental order in the system.

11.5) Post-conditions

The rental order information and status are displayed to the customer. The order data remains unchanged.

11.6) Extension Points

- Cancel Rental Order: The customer can cancel the rental order if its status is New.

12) Use Case: Cancel Rental Order

12.1) Brief Description

This use case allows customers to cancel a rental order when its status is New.

12.2) Flow of Events

12.2.1) Basic Flow

This use case begins when a customer selects the Cancel Rental Order function.

1. The system displays the selected rental order information.

2. The system checks the rental order status.
3. If the order status is New, the system displays a cancellation confirmation request.
4. The customer confirms the cancellation.
5. The system updates the rental order status to Cancelled.
6. The system displays a notification confirming the cancellation result.

12.2.2) Alternative Flows

- If the rental order is not in the New status, the system displays a notification indicating that the order cannot be cancelled.
- If the customer does not confirm the cancellation, the system keeps the original order status.

12.3) Special Requirements

A rental order can only be cancelled when its status is New.

12.4) Pre-conditions

The customer is logged in and has a rental order in the system.

12.5) Post-conditions

If successful, the rental order status is changed to Cancelled. Otherwise, the order status remains unchanged.

12.6) Extension Points

None.

13) Use Case: View Rental Orders

13.1) Brief Description

This use case allows staff members to view the list of rental orders in the system for monitoring and processing purposes.

13.2) Flow of Events

13.2.1) Basic Flow

This use case begins when a staff member selects the View Rental Orders function.

1. The system verifies the staff member's access permissions.
2. The system retrieves the list of rental orders.
3. The system displays the rental order information.
4. The staff member selects a rental order to view.
5. The system displays the detailed information of the selected order.

13.2.2) Alternative Flows

- If there are no rental orders, the system displays an empty list notification.
- If the staff member does not have permission to access the function, the system denies the request.

13.3) Special Requirements

No special requirements are specified.

13.4) Pre-conditions

The staff member is logged in to the system.

13.5) Post-conditions

The rental order list and detailed information are displayed. The rental order data remains unchanged.

13.6) Extension Points

- Update Rental Status: The staff member selects the function to update a rental order status.
- Manage Booking Records: The staff member selects the function to manage booking records.

14) Use Case: Update Rental Status

14.1) Brief Description

This use case allows staff members to update the processing status of rental orders during the rental service process.

14.2) Flow of Events

14.2.1) Basic Flow

This use case begins when a staff member selects the Update Rental Status function.

1. The system displays the list of rental orders.
2. The staff member selects the rental order to update.
3. The system displays the order information and its current status.
4. The staff member selects a new status according to the processing progress.
5. The staff member submits the update request.
6. The system validates the update information.
7. The system saves the new status and displays a confirmation message.

14.2.2) Alternative Flows

- If the update information is invalid, the system displays an error message and asks the staff member to check it.
- If the staff member cancels the operation, the system keeps the original order status.

14.3) Special Requirements

The system supports the following statuses: New, Confirmed, Delivering, In-Rental, Completed, and Cancelled.

14.4) Pre-conditions

The staff member is logged in, and the rental order to be updated exists in the system.

14.5) Post-conditions

If successful, the rental order status is updated. Otherwise, the status remains unchanged.

14.6) Extension Points

None.

15) Use Case: Manage Booking Records

15.1) Brief Description

This use case allows staff members to manage booking records, including viewing and tracking information related to vehicle bookings.

15.2) Flow of Events

15.2.1) Basic Flow

This use case begins when a staff member selects the Manage Booking Records function.

1. The system displays the list of booking records.
2. The staff member selects a record to view.
3. The system displays the booking information, including customer details, vehicle information, rental duration, and order information.
4. The staff member reviews the booking record.
5. The system displays the retrieved record information.

15.2.2) Alternative Flows

- If no matching record is found, the system displays a notification.
- If the staff member does not have permission to access the function, the system denies the request.

15.3) Special Requirements

No special requirements are specified.

15.4) Pre-conditions

The staff member is logged in to the system.

15.5) Post-conditions

The booking record information is displayed. The record data remains unchanged unless an update operation is performed.

15.6) Extension Points

- View Rental Orders: The staff member can access the rental order list.
- Update Rental Status: The staff member can access the function to update rental order statuses.

16) Use Case: Hand Over Vehicle

16.1) Brief Description

This use case allows staff members to hand over vehicles to customers according to confirmed rental order information.

16.2) Flow of Events

16.2.1) Basic Flow

This use case begins when a staff member selects the Hand Over Vehicle function.

1. The system displays the list of rental orders awaiting vehicle handover.
2. The staff member selects the corresponding rental order.
3. The system displays the customer, recipient, and vehicle information.

4. The staff member verifies the handover information.
5. The staff member hands over the vehicle to the customer.
6. The staff member confirms that the handover is complete.
7. The system updates the rental order status according to the rental process.

16.2.2) Alternative Flows

- If the rental order information is invalid or the order is not eligible for handover, the system displays a notification and does not record the handover.
- If the staff member cancels the operation, the system does not update the handover information.

16.3) Special Requirements

No special requirements are specified.

16.4) Pre-conditions

The staff member is logged in and has a rental order eligible for vehicle handover.

16.5) Post-conditions

If successful, the handover information is recorded and the rental order status is updated accordingly.

16.6) Extension Points

- Update Rental Status: The rental order status is updated after the vehicle handover.

17) Use Case: Receive Returned Vehicle

17.1) Brief Description

This use case allows staff members to receive vehicles returned by customers after the rental period ends.

17.2) Flow of Events

17.2.1) Basic Flow

This use case begins when a staff member selects the Receive Returned Vehicle function.

1. The system displays the list of rental orders currently in the In-Rental status.
2. The staff member selects the rental order associated with the returned vehicle.
3. The system displays the customer and vehicle information.
4. The staff member receives the vehicle from the customer.
5. The staff member inspects the condition of the returned vehicle.
6. The staff member records the inspection result.
7. The system saves the return information and updates the rental order status accordingly.

17.2.2) Alternative Flows

- If the corresponding rental order cannot be found, the system displays a notification indicating that the vehicle cannot be received.
- If the vehicle has not been returned, the staff member cannot complete the receiving process.

17.3) Special Requirements

The vehicle return process must be associated with the corresponding rental order.

17.4) Pre-conditions

The staff member is logged in, and the rental order is currently in the rental process.

17.5) Post-conditions

If successful, the vehicle receiving information and inspection results are saved in the system.

17.6) Extension Points

- Update Rental Status: The rental order status is updated after the vehicle is received.

18) Use Case: Manage Vehicles

18.1) Brief Description

This use case allows administrators to manage vehicle information and vehicle categories in the rental system.

18.2) Flow of Events

18.2.1) Basic Flow

This use case begins when an administrator selects the Manage Vehicles function.

1. The system verifies the administrator's access permissions.
2. The system displays the list of vehicles.
3. The administrator can choose to add a new vehicle.
4. If the administrator selects Update, the system displays the vehicle information to be edited.
5. If the administrator selects Delete, the system asks for confirmation.
6. The system validates the information and saves the changes.
7. The system displays a notification about the operation result.

18.2.2) Alternative Flows

- If the vehicle information is invalid, the system displays an error message and asks the administrator to correct it.
- If the administrator does not confirm the deletion, the system keeps the vehicle information unchanged.
- If the vehicle does not exist, the system displays a notification.

18.3) Special Requirements

Vehicle management includes managing vehicle categories, rental pricing models, and vehicle availability schedules.

18.4) Pre-conditions

The administrator is logged in to the system.

18.5) Post-conditions

If successful, vehicle information is added, updated, or deleted according to the administrator's request. Otherwise, the data remains unchanged.

18.6) Extension Points

None.

19) Use Case: Manage Customer Account

19.1) Brief Description

This use case allows administrators to manage customer accounts in the vehicle rental system.

19.2) Flow of Events

19.2.1) Basic Flow

This use case begins when an administrator selects the Manage Customer Account function.

1. The system displays the list of customer accounts.
2. The administrator selects an account to manage.
3. The system displays the customer's account information.
4. The administrator can add a new customer account.
5. The administrator can update account information.
6. The administrator can change account permissions or lock and unlock the account.
7. The system validates the information and saves the changes.
8. The system displays a notification about the operation result.

19.2.2) Alternative Flows

- If the account information is invalid, the system displays an error message.
- If the account does not exist, the system displays a notification.
- If the administrator cancels the operation, the system does not save the changes.

19.3) Special Requirements

Only administrators are allowed to manage customer accounts.

19.4) Pre-conditions

The administrator is logged in to the system.

19.5) Post-conditions

If successful, the customer account information or status is updated. Otherwise, the data remains unchanged.

19.6) Extension Points

None.

20) Use Case: Manage Order

20.1) Brief Description

This use case allows administrators to manage rental orders in the system, including viewing and updating rental order information.

20.2) Flow of Events

20.2.1) Basic Flow

This use case begins when an administrator selects the Manage Order function.

1. The system displays the list of rental orders.
2. The administrator selects a rental order to manage.
3. The system displays detailed order information, including customer details, vehicle information, rental duration, cost, and status.
4. The administrator updates the order information or status.
5. The administrator submits the update request.
6. The system validates the information.
7. The system saves the changes and displays a confirmation message.

20.2.2) Alternative Flows

- If the rental order does not exist, the system displays a notification.
- If the updated information is invalid, the system displays an error message and asks the administrator to check it.
- If the administrator cancels the operation, the system keeps the original rental order information.

20.3) Special Requirements

Only administrators are allowed to manage rental orders.

20.4) Pre-conditions

The administrator is logged in to the system.

20.5) Post-conditions

If successful, the rental order information is updated in the system. Otherwise, the data remains unchanged.

20.6) Extension Points

None.

21) Use Case: Manage Staff Account

21.1) Brief Description

This use case allows administrators to manage staff accounts in the vehicle rental system.

21.2) Flow of Events

21.2.1) Basic Flow

This use case begins when an administrator selects the Manage Staff Account function.

1. The system displays the list of staff accounts.
2. The administrator selects an account to manage.
3. The system displays the staff account information.
4. The administrator can add a new staff account.
5. The administrator can update account information.
6. The administrator can assign permissions or lock and unlock the staff account.

7. The system validates the information.
8. The system saves the changes and displays a confirmation message.

21.2.2) Alternative Flows

- If the account information is invalid, the system displays an error message and asks the administrator to correct it.
- If the staff account does not exist, the system displays a notification.
- If the administrator cancels the operation, the system does not save the changes.

21.3) Special Requirements

Only administrators are allowed to manage staff accounts and assign system permissions.

21.4) Pre-conditions

The administrator is logged in to the system.

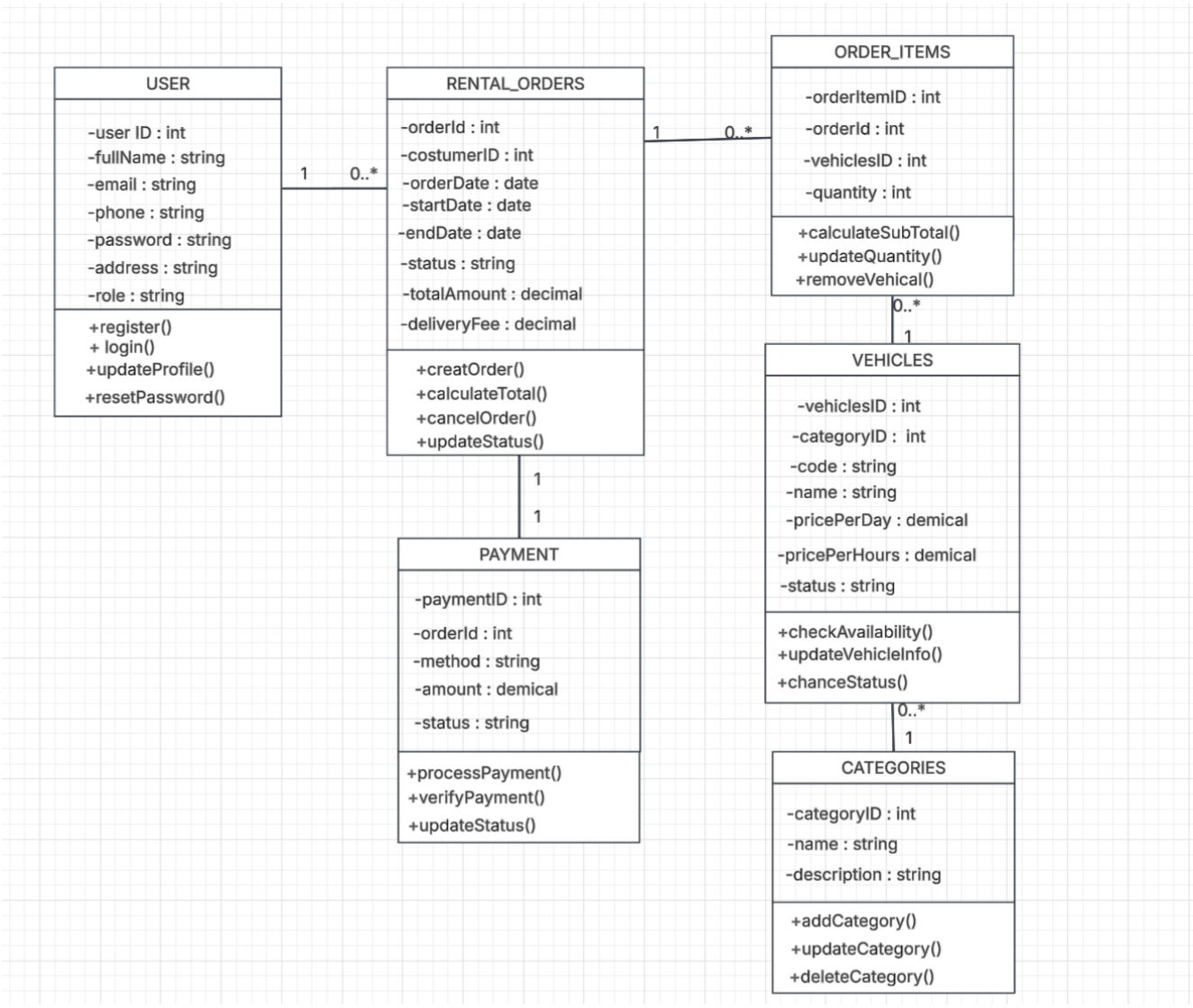
21.5) Post-conditions

If successful, the staff account is added, updated, or its status is changed according to the administrator's request. Otherwise, the data remains unchanged.

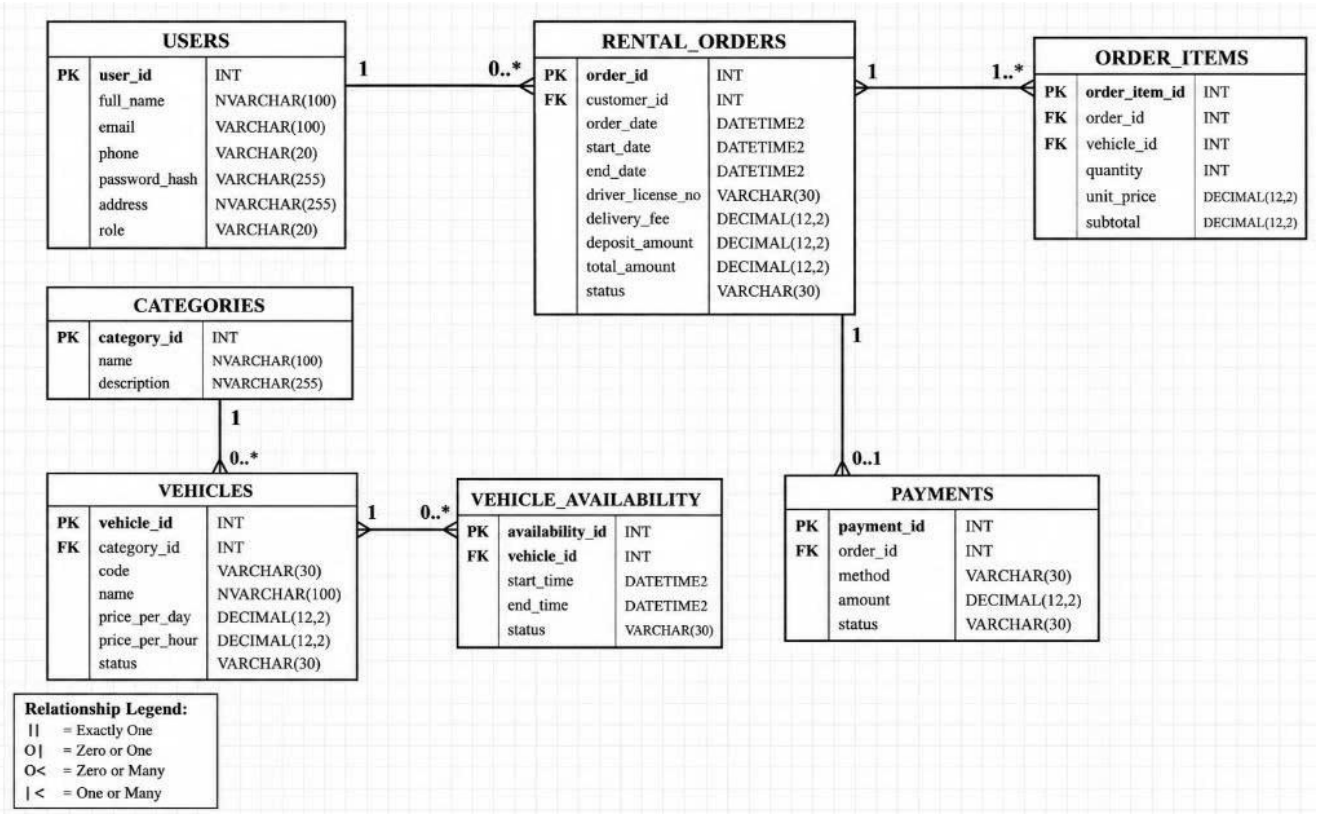
21.6) Extension Points

None.

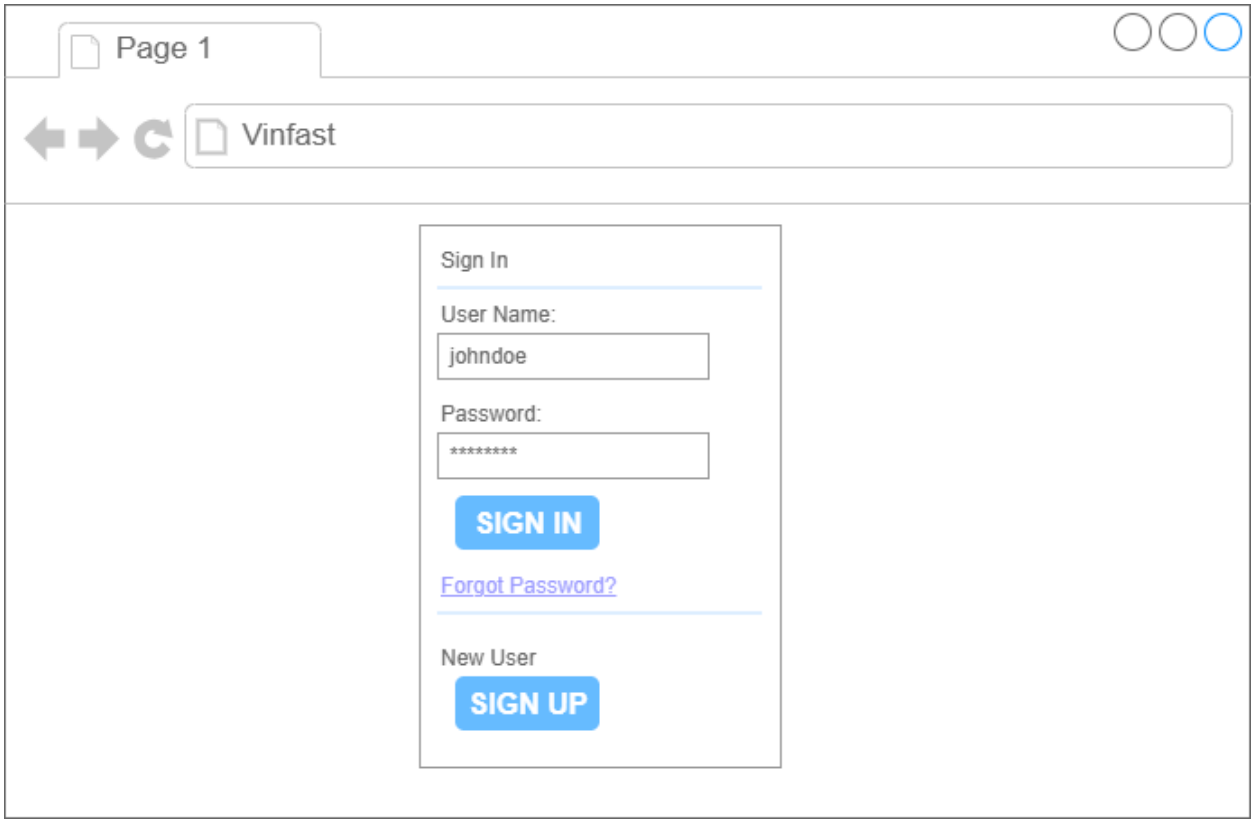
VI. Class Diagram



VII. Data Model




Page 1
Nhập mã xe và tên xe cần tìm
Trang chủ | Loại xe | Bảng giá | Hướng dẫn
Giỏ Xe
Đăng Nhập
Bộ Lọc Tìm Kiếm
Tất Cả Xe
Xe máy
Ô tô 4 chỗ
Ô tô 7 chỗ
Ảnh xe
VinFast VF8 - EV01
Giá: 1.200.000 VNĐ/ngày
Thuê Ngay
Ảnh xe
VinFast VF9
Giá: 1.200.000 VNĐ/ngày
Thuê Ngay



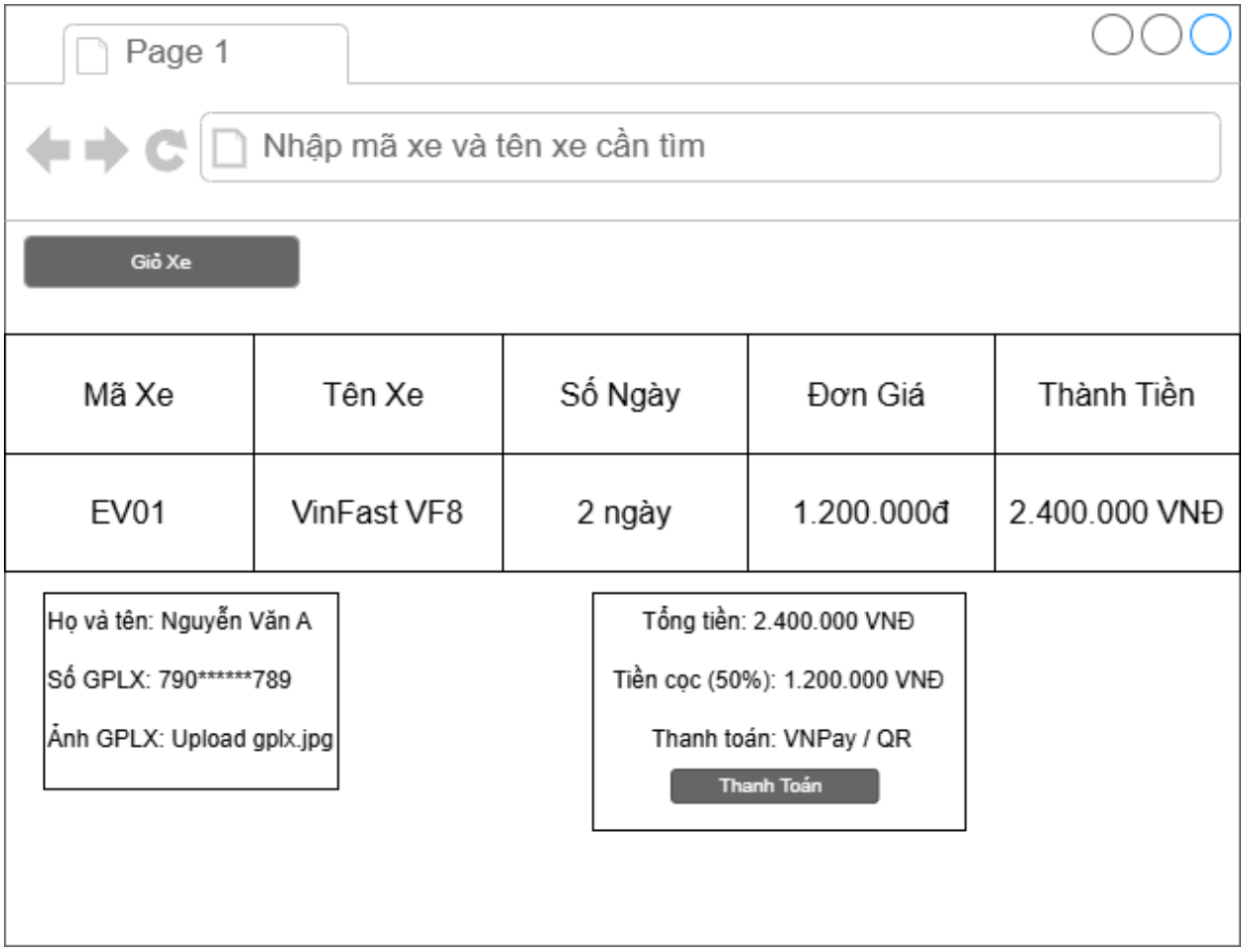
Functional Description:

- FR-01 (Display Vehicle List): The interface displays a catalog of VinFast electric vehicles (such as VF8 cars, Feliz S e-scooters) using a product card layout, showing thumbnail images, vehicle names, and daily rental rates.
- FR-02 (Vehicle Search): A search bar enables users to enter vehicle IDs or vehicle names for instant filtering.
- FR-03 (Filter and Category): Category filters (Electric Cars vs. Electric Scooters) and price ranges help customers easily find vehicles matching their specific needs.
- FR-04 (User Authentication and Navigation):
    - The top navigation bar provides quick access to Home, Pricing, and User Guide sections.
    - A Login / Register button is integrated on the top right. Clicking this button opens a login modal requesting the user's username (or phone number / email) and password.
    - Upon successful authentication, the system automatically checks user role permissions (Customer, Staff, or Admin) and redirects to the appropriate interface.



Functional Description:

- FR-05 (View Vehicle Specifications): Displays high-resolution vehicle imagery alongside detailed technical specifications (vehicle line, battery capacity, driving range, transmission type) and daily rental pricing.
- FR-07 (Rental Duration Selection): Customers select pickup and return dates using an interactive calendar picker. The system automatically calculates total rental days and enables the "Add to Cart" button.



Functional Description:

- FR-06 (Driver's License Verification): Customers input personal details (Full Name, License Number) and upload photos of their driving license for data verification.
- FR-08 (Online Deposit Payment): An order summary table lists Vehicle ID, rental duration, unit price, and total cost. Customers confirm a 50% deposit payment and select a payment gateway (VNPay or QR Code) to finalize the reservation.



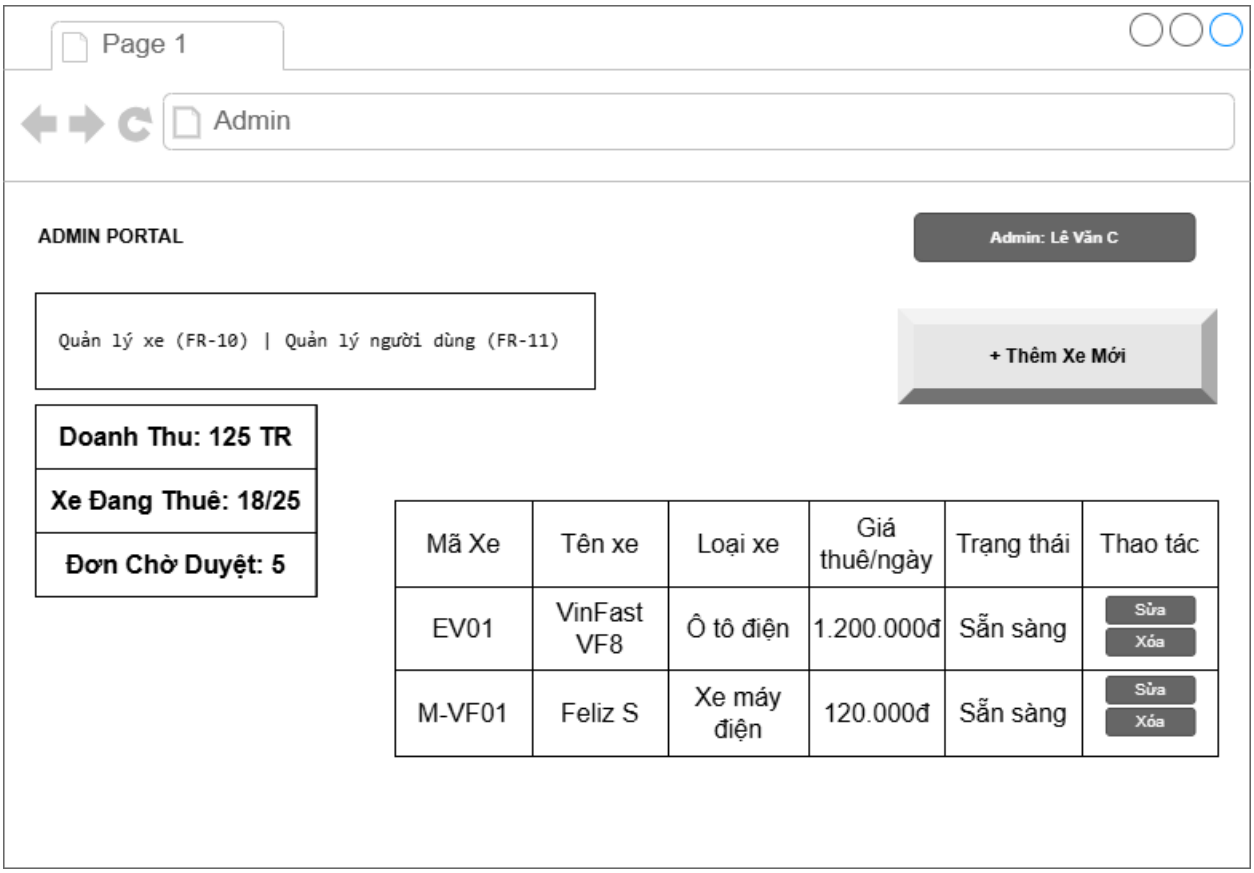
Functional Description:

- FR-09 (Track and Cancel Orders): Allows users to review their complete personal booking history along with processing statuses such as Pending, New Order, or Completed.
- Provides quick status filter tabs and allows customers to manually cancel bookings for orders that are still pending vehicle handover.



Functional Description:

- FR-12 (Staff Handover Workflow): A dedicated portal for operation staff to execute vehicle pickup and return procedures at the station.
- Staff members search for booking IDs, complete safety checklist items (battery level > 90%, exterior condition check, insurance documents), and input the current odometer reading before clicking "Confirm Vehicle Handover Complete".



Functional Description:

- FR-10 (Fleet Management and Pricing): Enables administrators to view all vehicles in the system, update status (Available or Rented), edit pricing, or add new vehicles to the fleet.
- FR-11 (User Account Management and Analytics): Provides quick operational metric cards (Total Revenue, Active Rentals, Pending Approvals) and manages user and staff role permissions.